

# HAT-P-36b

R-1842

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-36\_160326 · created 2026-08-03T07:06:42+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457473.9087 +/- 0.0013 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.092 +/- 0.023                 |
| Transit depth (Rp/Rs)^2                           | 0.84 +/- 0.42 [%]               |
| Orbital Inclination (inc)                         | 89.58 +/- 2.59                  |
| Airmass coefficient 1 (a1)                        | 0.988 +/- 0.02                  |
| Airmass coefficient 2 (a2)                        | 0.009 +/- 0.013                 |
| Scatter in the residuals of the lightcurve fit is | 2.06 %                          |
| Best Comparison Star                              | #2 - [220, 370]                 |
| Optimal Aperture                                  | 3.0                             |
| Optimal Annulus                                   | 9.73                            |
| Transit Duration (day)                            | 0.0932 +/- 0.0032               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.092 $\pm$ 0.023 vs 0.126 (deviation 1.05 $\sigma$ ) |
| Duration fit vs published            | 0.0932 d vs 0.0925 d (deviation 0.16 $\sigma$ )       |
| Mid-time O-C                         | 3.2 min                                               |
| Depth SNR                            | 2.8                                                   |
| Residual scatter                     | 2.06 %                                                |

### Light curve

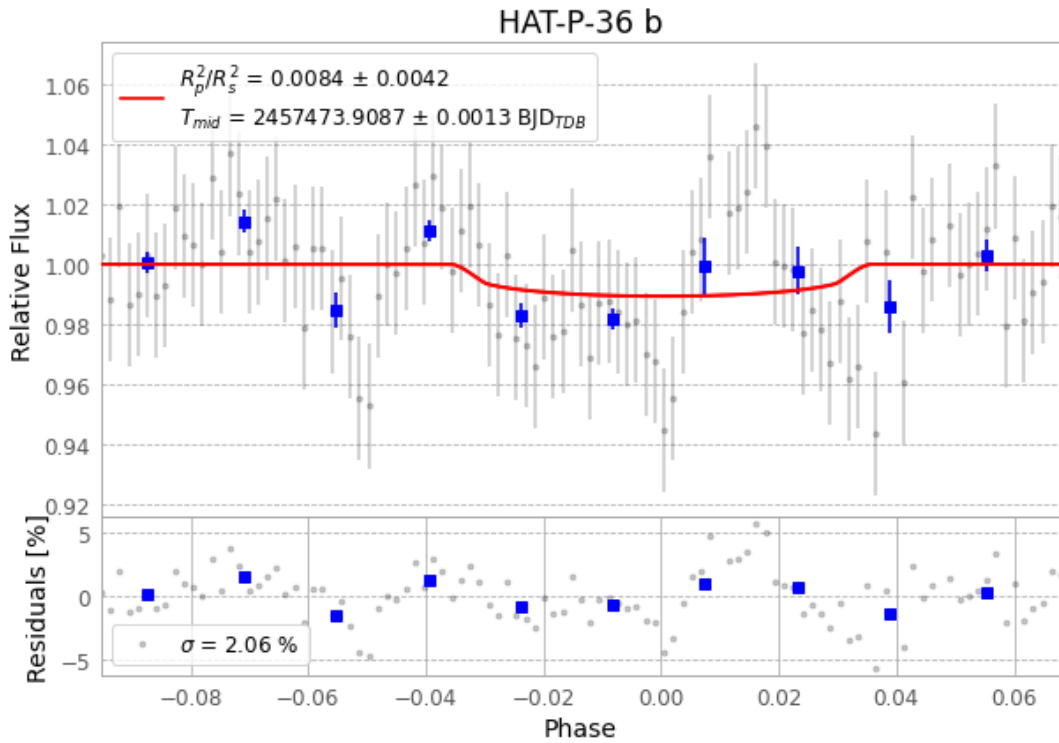


Figure 1 — FinalLightCurve\_HAT-P-36 b\_2016-03-25.png (SHA-256 29a8f026e45d0a08...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [361, 225], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 e580fb2b37242229...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

105 FITS frames (69 MB), HATP-36160326063912.FITS → HATP-36160326115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-160326.log (df66504bc574...), FinalLightCurve\_HAT-P-36 b\_2016-03-25.png (29a8f026e45d...), FinalLightCurve\_HAT-P-36 b\_2016-03-25.csv (a0835895fac0...)

## Compute resources

Wall time 195.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 07:06Z.